Description

`num` and `den` are the numerator and denominator of the fraction `N/D` reduced to lowest terms. `den` is always positive. If `N` and `D` are the same sign, `num` is positive; otherwise `num` is negative.

Examples

```
ratio<4,6>::num == 2
ratio<4,6>::den == 3
ratio<4,-6>::num == -2
ratio<4,-6>::den == 3
```

D.6.2. std::ratio_add template alias

The `std::ratio_add` template alias provides a mechanism for adding two `std::ratio` values at compile time, using rational arithmetic.

Definition

```
template <class R1, class R2>
using ratio_add = std::ratio<see below>;
```

Preconditions

`R1` and `R2` must be instantiations of the `std::ratio` class template.

Effects

`ratio_add<R1,R2>` is defined as an alias for an instantiation of `std::ratio` that represents the sum of the fractions represented by `R1` and `R2` if that sum can be calculated without overflow. If the calculation of the result overflows, the program is ill formed. In the absence of arithmetic overflow, `std::ratio_add<R1,R2>` shall have the same `num` and `den` values as `std::ratio<R1::num * R2::den + R2::num * R1::den, R1::den * R2::den>`.